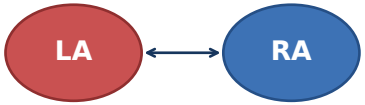
A Chamber-Aware Computational Framework Reveals Matrix–Metabolic Remodeling in Atrial Fibrillation

1 Frozen healthy reference

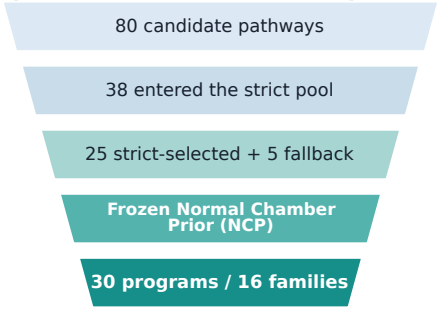
Healthy paired LA/RA reference



8 healthy donors with paired chambers

Donor-level LA–RA contrast
align to healthy direction → scale by healthy magnitude

Prespecified freeze and selection provenance



22 LA-biased | 8 RA-biased programs

Healthy calibration — not AF discovery

2 Cross-dataset deployment

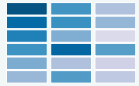
A Paired-chamber bulk deployment

3 cohorts

25 AF / 26 SR donors

Frozen NCP;
no pathway reselection

Selective cross-cohort remodeling



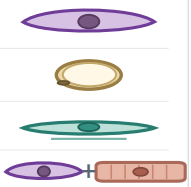
High 6 | Moderate 8 | Low 2

High consistency spans LA translation/RQC and RA matrix/proteoglycan/developmental families.

B LA single-nucleus localization

18 AF / 16 controls · 9 LA compartments ·
4/16 families non-Below-detection

F10	Fidelity → fibroblast
F11	Relay → adipocyte
F15	Relay → endothelial
F16	Extension → fibroblast + atrial cardiomyocyte



Schematic cells; nominal LA-only localization—not lineage or FDR-controlled discovery.

C Cross-sectional AF-duration context

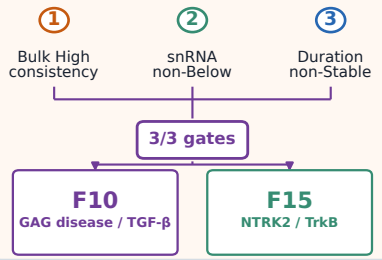
12 AF-LAA donors · family activity vs. duration

Stable	Weak	Strong
8 families	**6** families	**2** families **F2, F9**
$\lvert\rho_{LAA}\rvert < 0.2$	$0.2 \le \lvert\rho_{LAA}\rvert < 0.4$	$\lvert\rho_{LAA}\rvert \ge 0.4$

Classes use full-precision cross-sectional $\lvert\rho_{LAA}\rvert$; not longitudinal follow-up, recurrence, or treatment response.

3 Prioritization and context

Path A: all three transcriptomic gates satisfied



Spatial context (GeoMx)

24 eligible ROIs from 4 patients

ROI-level summaries only

No patient-clustered inference

Not a Path A gate

Genetic calibration and traceability

Local AF GWAS: 7/12 anchors

Atrial eGene: 1/12 (SMAD9)

PP3-dominant: 12/12

PP4 ≥ 0.70: 0/12

Non-nested; not a gate or causal claim

A frozen chamber-aware reference enables consistent cross-dataset comparison and assigns F10 and F15 to Path A across three transcriptomic layers.

Spatial and genetic analyses provide context and calibration; they do not establish patient-level confirmation or causal mechanisms.

AF, atrial fibrillation; SR, sinus rhythm; LA/RA, left/right atrium; RQC, ribosome quality control; ROI, region of interest.